

A Survey of Large Audio Language Models: Generalization, Trustworthiness, and Outlook

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Abstract: The foundational capabilities established by Large Language Models (LLMs) have paved the way for Multimodal Large Language Models (MLLMs), within which Large Audio Language Models (LALMs) are essential for realizing universal auditory intelligence. Despite their remarkable performance, the escalation of LALMs' capabilities has significantly outpaced the development of systemic frameworks to ensure their trustworthiness. This survey provides a comprehensive investigation into the endogenous mechanisms of LALMs, detailing the architectural innovations and alignment algorithms that facilitate emergent reasoning. Specifically, we analyze how the transition to unified end-to-end frameworks and the integration of continuous acoustic signals inherently expand the attack surface. To rigorously evaluate the risks within these paradigms, we establish a comprehensive taxonomy of trustworthiness, categorizing critical vulnerabilities such as cross-modal jailbreaking, latent acoustic backdoors, and biometric privacy leakage. We review the state-of-the-art through six analytical pillars: hallucination, robustness, safety, privacy, fairness, and authentication. The profound imbalance between a mature offensive landscape and underdeveloped defenses further validates the critical trustworthiness gaps and multidimensional risks facing audio-centric intelligence. Finally, we propose a strategic roadmap advocating for "Defense-in-Depth" architectures, causal auditory world modeling, and intrinsic representation engineering to bridge the gap between empirical performance and intrinsically trustworthy audio intelligence.

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🔗 Project: <https://github.com/Kwwww74/Awesome-Trustworthy-AudioLLMs>